LECTURE 11: NORMALISATION

COMP2013J: Databases and Information Systems

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Normal Form and Normalisation?

- A normal form is a property of a relational database
- When a relation is non-normalized (that is, does not satisfy a normal form), then it presents **redundancies** and produces **undesirable behaviour** during update operations.

- For the schemas that do not satisfy a normal form, we can apply a procedure known as **normalization**.
- Normalization allows the non-normalized schemas to be transformed into new schemas for which the satisfaction of a normal form is guaranteed.

Normal Forms

- Basically, the normal form of the data indicates how much redundancy is in the data.
- The normal forms have a strict ordering:
  - 1st Normal Form
  - 2nd Normal Form
  - 3rd Normal Form
  - Boyce-Codd Normal Form (BCNF)
  - 4th Normal Form
  - 5th Normal Form

Problem of Redundancy

- If redundancy exists then this can cause problems during normal database operations:
  - When data is inserted into the database, the data must be duplicated wherever redundant versions of that data exists.
  - When data is updated, all redundant data must be updated at the same time.

Problem of Redundancy

The key is made up of the attributes Employee and Project

Employee	Salary	Project	Budget	Function
John Brown	20000	Mars	2,000,000	Technician
Paul Green	35000	Jupiter	15,000,000	Designer
Paul Green	35000	Venus	15,000,000	Designer
Fred Hoskins	55000	Venus	15,000,000	Manager
Fred Hoskins	55000	Jupiter	15,000,000	Consultant
Fred Hoskins	55000	Mars	2,000,000	Consultant
Martin Moore	48000	Mars	2,000,000	Manager
Martin Moore	48000	Venus	15,000,000	Designer
Ross Kemp	48000	Venus	15,000,000	Designer
Ross Kemp	48000	Jupiter	15,000,000	Manager

Where do we see **redundancy** in this table?

Functional Dependencies

- Sometimes the starting point for understanding the data requirements is given using **functional dependencies**.
- A function dependency is shown by two lists of attributes separated by an arrow.
  - If we know the values for the left-side attributes, we can find the values for the right.
- **Example**, a relation: student(s_number, firstname, lastname, advisor_number, advisor_name)
  - s_number → firstname, lastname, advisor_number, advisor_name
  - This is a functional dependency: if we know somebody's student number, we can find their name, their advisor's number and their advisor's name.

Functional Dependencies

- student(s_number, firstname, lastname, advisor_number, advisor_name)

- In the same way:
  - advisor_number → advisor_name
  - This is also a functional dependency: if we know the ID number of a student’s advisor, we can find their name.

- But:
  - lastname, firstname →s_number
  - advisor_name → advisor_number
  - These are **NOT** functional dependencies. Two students could have the same name, so we cannot reliably find the student’s ID number from a name. Similarly, two advisors could have the same name.

1ST NORMAL FORM

Example Relation

- student(<u>Snum</u>, Name, DoB, (Subject, Grade))

<u>SNum</u>	**Name**	**DoB**	**Subject**	**Grade**
12345	Smith.J	02/03/96	Java	B
			Soft Eng	C
			Databases	A
23456	White.A	04/02/94	Java	D
			Soft Eng	B
34567	Moore.T	06/10/95	Databases	A
			Soft Eng	B
			Networks	C
45678	Smith.J	02/11/98	Soft Eng	C

Example Relation

- student(<u>Snum</u>, Name, DoB, (Subject, Grade))

<u>SNum</u>	Name	DoB	Subject	Grade
12345	Smith.J	02/03/96	Java	B
			Soft Eng	C
			Databases	A
23456	White.A	04/02/94	Java	D
			Soft Eng	B
	Moore.T	06/10/95	Databases	A
			Soft Eng	B
			Networks	C
	Smith.J	02/11/98	Soft Eng	C

"subject" and "grade" make up a **repeating group**.

Each record has multiple entries for these attributes.

This is an "unnormalised" table (and can't actually be stored in a relational database management system).

First Normal Form (1NF)

- First Normal Form (1NF) deals with the shape of the record type.
- **A relation is in 1NF if, and only if, it contains no repeating attributes or groups of attributes.**
- **Example**:
  - The student table with the repeating group (subject, grade) is not in 1NF.
- To remove the repeating group, one of two things can be done:
  - ”Flatten” the relation (fill in the empty attribute spaces) and extend the key, or
  - “Decompose” the relation (divide into multiple relations)

Flatten and Extend Primary Key

- The original student table (with the repeating group) can be written as:
  - student(<u>Snum</u>, Name, DoB, (Subject, Grade))
- If the repeating group was flattened, it would look something like:
  - student(<u>Snum</u>, Name, DoB, <u>Subject</u>, Grade)
- This does not have repeating groups, but has redundancy. For every combination of SNum/Subject, the student's name and date of birth is duplicated. This can lead to errors known as **anomalies**.
- Three main types: **insertion**, **update** and **deletion** anomalies.

Insertion Anomaly

- As “subject” is now part of the primary key, we cannot add a student until they have at least one subject

This is not possible, because “subject” cannot be NULL.

<u>SNum</u>	**Name**	**DoB**	**<u>Subject</u>**	**Grade**
53342	Ford.P	2/5/99		
12345	Smith.J	02/03/96	Java	B
12345	Smith.J	02/03/96	Soft Eng	C
12345	Smith.J	02/03/96	Databases	A
23456	White.A	04/02/94	Java	D
23456	White.A	04/02/94	Soft Eng	B
34567	Moore.T	06/10/95	Databases	A
23456	White.A	04/02/94	Soft Eng	B
23456	White.A	04/02/94	Networks	C
45678	Smith.J	02/11/98	Soft Eng	C

Update Anomaly

- Changing the name of a student means finding all rows of the database where that student exists and changing each one separately.

Oops: we forgot to change this one, the data is not consistent anymore.

<u>SNum</u>	Name	DoB	<u>Subject</u>	Grade
12345	Smythe.J	02/03/96	Java	B
12345	Smythe.J	02/03/96	Soft Eng	C
12345	Smith.J	02/03/96	Databases	A
23456	White.A	04/02/94	Java	D
23456	White.A	04/02/94	Soft Eng	B
34567	Moore.T	06/10/95	Databases	A
23456	White.A	04/02/94	Soft Eng	B
23456	White.A	04/02/94	Networks	C
45678	Smith.J	02/11/98	Soft Eng	C

Deletion Anomaly

- Deleting details about one thing can also mean we delete something else.

If we delete all Soft Eng subject information, we lost all data about student 4567 also.

SNum	Name	DoB	Subject	Grade
12345	Smith.J	02/03/96	Java	B
12345	Smith.J	02/03/96	Soft Eng	C
12345	Smith.J	02/03/96	Databases	A
23456	White.A	04/02/94	Java	D
23456	White.A	04/02/94	Soft Eng	B
34567	Moore.T	06/10/95	Databases	A
23456	White.A	04/02/94	Soft Eng	B
23456	White.A	04/02/94	Networks	C
45678	Smith.J	02/11/98	Soft Eng	C

Decomposing the Relation

- The alternative approach is to split the table into two relations: one for the repeating groups and one for the non-repeating groups.
- The primary key for the original relation is included in both of the new relations.
- We can return to the original table by using a JOIN operation on these relations: **non-loss decomposition**.

No repeating groups: 1NF

Students

SNum	**Name**	**DoB**
12345	Smith.J	02/03/86
23456	White.A	04/02/84
34567	Moore.T	06/10/85
45678	Smith.J	02/11/88

Grades

SNum	**Subject**	**Grade**
12345	Databases	B
12345	VB	C
12345	Soft. Eng	A
23456	VB	D
…..	….	….
45678	Soft. Eng	C

2^{ND} NORMAL FORM

Second Normal Form (2NF)

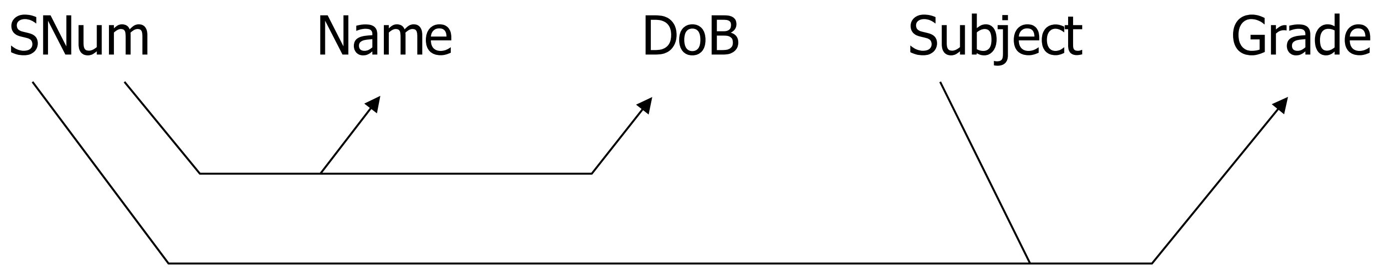
- **A relation is in 2NF if, and only if, it is in 1NF and every non-key attribute is fully functionally dependent on the whole key**.
- The relation must be in 1NF and all non-key attributes must depend on the whole key, not just part of it.
  - In other words: there must be no **partial key dependencies.**
- The problem arises when there is a **compound key**, e.g. in the Grades relation: <u>SNum</u>, <u>Subject</u>
- In this case it is possible for non-key attributes to depend on only part of the key (i.e. on only one of the key attributes).

2NF Example

- Consider the flattened student relation:
  - student(<u>SNum</u>, Name, DoB, <u>Subject</u>, Grade)
- There are no repeating groups: already in 1NF.
- However, there is a compound primary key, so we must check that the non-key attributes depend on the whole key.
  - SNum determines the Name and DoB but not Grade.
    - **SNum → Name, DoB**
  - Subject together with SNum determines Grade, but not Name or DoB
    - **SNum, Subject→ Grade**
- There is a problem with potential redundancies.

Dependency Diagram

- We can use a **dependency diagram** to show how non-key attributes relate to each part or combination of parts of the primary key.



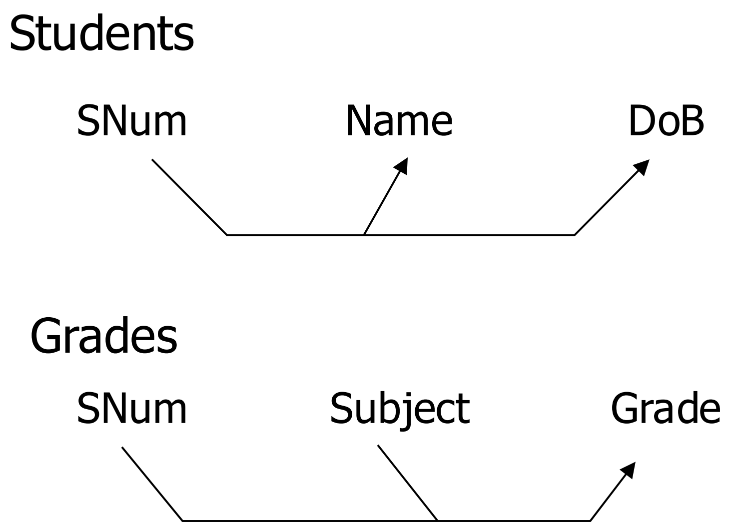
- This relation is not in 2NF. It appears to be two tables squashed into one.
- Solution: decompose the relation.

2NF

Procedure:

1. Create a new relation that contains all of the attributes that are solely dependent on SNum (SNum is the primary key)
2. Create a new relation that contains all the attributes that are solely dependent on Subject.
   - In this example, there are no attributes that depend only on Subject.
3. Create a new relation with all the attributes that depend on both SNum and Subject.
   - The primary key is <u>SNum</u>, <u>Subject</u>

2NF



- All attributes in each relation are fully functionally dependent on the primary key.
- Both relations are now in 2NF.

- Interesting: this is the same set of relations we got when we decomposed to remove the repeating group.

3RD NORMAL FORM

Third Normal Form (3NF)

- 3NF is an even stricter normal form that removes almost all redundant data.
- **A relation is in 3NF if, and only if, it is in 2NF and there are no transitive functional dependencies**.
- Transitive functional dependencies are when one non-key attribute is functionally dependent on another non-key attribute.
- By definition, a transitive functional dependency can only happen if there is more than one non-key attribute.
  - Any relation in 2NF with < 2 non-key attributes must automatically be in 3NF.

3NF: Example

Projects

Proj_No	Manager	Address
P1	Black. B	32 High St
P2	Smith. J	11 New St
P3	Black. B	32 High St
P4	Black. B	32 High St

- “Projects” has more than one non-key field (Manager and Address) so we must check for transitive dependency.

3NF: Example

- In this example, we are told that Address depends on the value of Manager.
  - If we know a project's manager, we can find the address.

- Projects(<u>Proj_No</u>, Manager, Address)
  - Manager → Address

- In this case, Address is **transitively dependent** on manager.

- The primary key is <u>Proj_No</u>, but the functional dependency makes no reference to this key.

3NF: Example

- Data redundancy can come from this:
  - We duplicate the address if a manager is in charge of more than one project.
  - Causes problems if we have to change the address, because it must be changed in several places.
- Solution: **decompose**:
  - Create two relations: one with the transitive dependency in it, and another for all of the remaining attributes.
  - Split Projects into Projects and Managers
  - In the Projects relation, we keep the same primary key: <u>Proj_no</u>
  - In the Managers relation we use the left side of the functional dependency as the primary key: <u>Manager</u>

3NF: Example Result

- Now we need to store the address only once.
- If we need to know a manager's address, we can look it up in the Managers relation.
- The "manager" attribute is the link between the two tables: in the Projects relation it is now a foreign key.
- These relations are now in 3NF

Projects

Proj_No	Manager
P1	Black.B
P2	Smith.J
P3	Black.B
P4	Black.B

Managers

Manager	Address
Black.B	32 High St
Smith.J	11 New St

Summary: 1NF

- A relation is in 1NF if it has no repeating groups.
- To convert an unnormalised relation to 1NF, either:
  - Flatten the table and extend the primary key or
  - Decompose the relation into smaller relation: one for the repeating groups and one for the non-repeating groups.
    - Remember to put the primary key from the original relation into both new relations.
- Decomposition often gives the best results:
  - R(a, b, (c, d)) becomes:
    - R(a, b)
    - R1(a, c, d)

Summary: 2NF

- A relation is in 2NF if it contains no repeating groups (1NF) and no partial key functional dependencies.
  - Rule: A relation in 1NF with a single key attribute must be in 2NF.
- To convert a relation with partial key functional dependencies to 2NF, create a new set of relations:
  - One relation for the attributes that are fully dependent on the key.
  - One relation for each part of the key that has partially dependent attributes.
- R(<u>a</u>, <u>b</u>, c, d) and a→c becomes:
  - R(<u>a</u>, <u>b</u>, d)
  - R1(<u>a</u>, c)

Summary: 3NF

- A relation is in 3NF if it contains no repeating groups (1NF), no partial key functional dependencies (2NF), and no transitive functional dependencies.
- To convert a relation with transitive functional dependencies to 3NF, remove the attributes involved in the transitive dependency and put them in a new relation.
  - Rule: a relation in 2NF with only one non-key attribute must be in 3NF.
- In a normalised relation, a non-key field must provide data about "the key, the whole key and nothing but the key".

BOYCE-CODD NORMAL FORM

Boyce-Codd Normal Form

- Boyce-Codd Normal Form (BCNF) is named after Raymond Boyce and Edgar Codd who developed it in 1974 to address types of anomaly not addressed in 3NF.
  - Sometimes referred to as 3.5NF
- BCNF relies on the concept of a **candidate key**.
- A candidate key is an attribute (or group of attributes) that are capable of uniquely identifying any row in a relation.
  - In other words, attributes that could be used as a primary key.

BCNF: Candidate Keys

- Frequently, a relation has only one candidate key, which is used as the primary key.
- However, there are sometimes multiple candidate keys. When this happens, we choose one to be the primary key.
- For example, a relation named “Students” that contains the following attributes:
  - `UCD_ID`
  - `BJUT_ID`
- Both are candidate keys, as they uniquely identify a student, though we choose only one to be the primary key.

BCNF

- When a relation has more than one candidate key, anomalies can result even though the relation is in 3NF.
- 3NF does not deal with **overlapping candidate keys**.
  - i.e. composite candidate keys with at least one attribute in common.
- BCNF is based on the concept of a **determinant**.
  - A determinant is any attribute (or group of attributes) that some other attribute is fully functionally dependent on (i.e. the left side of a functional dependency).
- A relation is in BCNF if, and only if, every determinant is a candidate key.

BCNF: Hospital Example

PatNo	PatName	AppSlot	Time	Doctor
1	Eamonn	0	09:00	Octopus
2	Eoin	0	09:00	Evil
3	Arnold	1	10:00	Octopus
4	Stephen	0	13:00	Evil
5	Patricia	1	14:00	Octopus

- **Extra information:**
  - Every patient has a unique patient number.
  - Patients with names beginning with a letter before ‘P’ get morning appointments.
  - The Appointment slots start at 0 for the first appointment of the morning or afternoon, 1 for the second and so on.

BCNF: Hospital Example

- Appointments(PatNo, PatName, AppSlot, Time, Doctor)
- We are given some functional dependencies (mostly based on the extra information):
  - PatNo → PatName
  - PatNo, AppSlot → Time, Doctor
  - Time → AppSlot
- We have two options for selecting a primary key:
  - Appointments(PatNo, PatName, AppSlot, Time, Doctor): example A
  - Appointments(PatNo, PatName, AppSlot, Time, Doctor): example B

BCNF: Example A

- Appointments(PatNo, PatName, AppSlot, Time, Doctor)
- No repeating groups, so in 1NF.
- 2NF – eliminate partial key dependencies:
  - Appointments(PatNo, AppSlot, time, Doctor)
  - Patients(PatNo, PatName)
- 3NF – no transitive dependencies so it's already in 3NF.
- Now try BCNF.

BCNF: Every determinant must be a candidate key.

- Appointments(<u>PatNo</u>, <u>AppSlot</u>, time, Doctor)
- Patients(<u>PatNo</u>, PatName)

- For Appointments: is determinant a candidate key for each of the functional dependencies?

- PatNo → PatName
  - PatNo is present in Appointments, but not PatName, so this is not relevant.

BCNF: Every determinant must be a candidate key.

- Appointments(<u>PatNo</u>, <u>AppSlot</u>, time, Doctor)
- Patients(<u>PatNo</u>, PatName)

- For Appointments: is determinant a candidate key for each of the functional dependencies?

- PatNo, AppSlot → Time, Doctor
  - All of these attributes are present in Appointments, so this functional dependency is relevant. Is this a candidate key?
  - PatNo, AppSlot **is** the key, so this is a candidate key: OK.

BCNF: Every determinant must be a candidate key.

- Appointments(<u>PatNo</u>, <u>AppSlot</u>, time, Doctor)
- Patients(<u>PatNo</u>, PatName)

- For Appointments: is determinant a candidate key for each of the functional dependencies?

- Time → AppSlot
  - Time is present in Appointments and so is AppSlot, so it's relevant.
  - If this is a candidate key, then we could rewrite Appointments as:
    - Appointments(PatNo, Appslot, <u>Time</u>, Doctor).
  - This won't work, so it is not in BCNF. "Time" is not a candidate key.

Rewrite to BCNF

- Appointments(PatNo, AppSlot, time, Doctor)
- Patients(PatNo, PatName)

- Rewrite to BCNF:
  - Appointments(PatNo, Time, Doctor)
  - Patients(PatNo, PatName)
  - Slots(Time, AppSlot)

- “Time” is enough to work out the appointment slot of a patient.
- Now BCNF is satisfied, and the final relations shown are in BCNF.

BCNF: Example B

- Appointments(PatNo, PatName, AppSlot, Time, Doctor)
- No repeating groups, so it's in 1NF
- 2NF – eliminate partial key dependencies:
  - Appointments(PatNo, Time, Doctor)
  - Patient(PatNo, PatName)
  - Slots(Time, AppSlot)
- 3NF – no transient dependencies, so it's in 3NF
- Now try BCNF.

BCNF Check

- Appointments(<u>PatNo</u>, <u>Time</u>, Doctor)
- Patient(<u>PatNo</u>, PatName)
- Slots(<u>Time</u>, AppSlot)

- For Appointments: is determinant a candidate key for each of the functional dependencies?

- PatNo → PatName
  - PatNo is present in Appointments, but no PatName, so it’s not relevant.

BCNF Check

- Appointments(PatNo, Time, Doctor)
- Patient(PatNo, PatName)
- Slots(Time, AppSlot)

- For Appointments: is determinant a candidate key for each of the functional dependencies?

- PatNo, AppSlot → Time, Doctor
  - PatNo and AppSlot are not both present in Appointmetns, so this is not relevant.

BCNF Check

- Appointments(<u>PatNo</u>, <u>Time</u>, Doctor)
- Patient(<u>PatNo</u>, PatName)
- Slots(<u>Time</u>, AppSlot)

- For Appointments: is determinant a candidate key for each of the functional dependencies?

- Time → AppSlot
  - Time is present in Appointments, but not AppSlot, so this is not relevant.
- Relations are in BCNF

Example Summary

This example demonstrates three things:

1. BCNF is stronger than 3NF: relations in 3NF are not necessarily in BCNF.
2. BCNF is needed in certain situations to get a full understanding of the data model.
3. There are several routes to take to arrive at the same set of relations in BCNF.
   - Unfortunately there are no rules to guarantee the easiest route.

What problem does BCNF overcome?

Student_No	Major	Supervisor
123	Physics	Einstein
123	Music	Mozart
456	Biology	Darwin
789	Physics	Bohr
999	Physics	Einstein

- We have the following functional dependencies:
  - Student_No, Major → Supervisor
  - Supervisor → Major

What problem does BCNF overcome?

Student_No	**Major**	**Supervisor**
123	Physics	Einstein
123	Music	Mozart
456	Biology	Darwin
789	Physics	Bohr
999	Physics	Einstein

- No repeating groups, so it's in 1NF
- No partial key dependencies, so it's in 2NF
- There's only one non-key attribute (Supervisor) so it must be in 3NF.

What problem does BCNF overcome?

Student_No	**Major**	**Supervisor**
123	Physics	Einstein
123	Music	Mozart
456	Biology	Darwin
789	Physics	Bohr
999	Physics	Einstein

- If the record for student 456 is deleted we lose not only information about that student, but also the fact that Darwin advises in Biology.
- We cannot record the fact that Watson can advise on Computing until we have a student doing a project on Computing that has Watson as an advisor.

What problem does BCNF overcome?

- In BCNF we have two tables, and these problems are eliminated:

Student_No	**Supervisor**
123	Einstein
123	Mozart
456	Darwin
789	Bohr
999	Einstein

Supervisor	**Major**
Einstein	Physics
Mozart	Music
Darwin	Biology
Bohr	Physics

Summary

- A relation is in 1NF if, and only if, it contains no repeating groups.
- A relation is in 2NF if, and only if, it is in 1NF and every non-key attribute is fully functionally dependent on the whole key.
- A relation is in 3NF if, and only if, it is in 2NF and has no transitive functional dependencies.
- A relation is in BCNF if, and only if, it is in 3NF and every determinant is a candidate key.